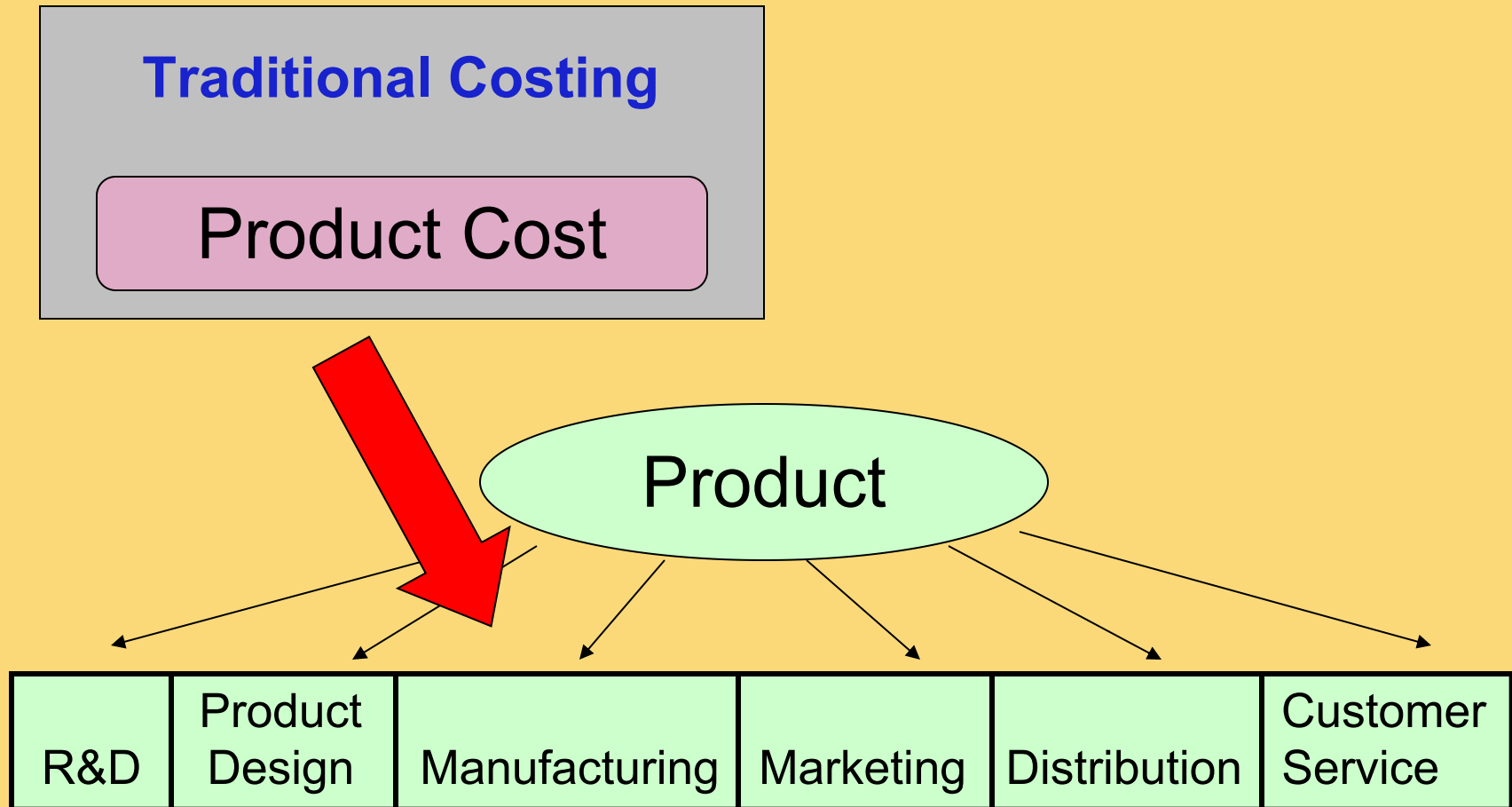


Activity-Based Costing: A Tool to Aid Decision Making

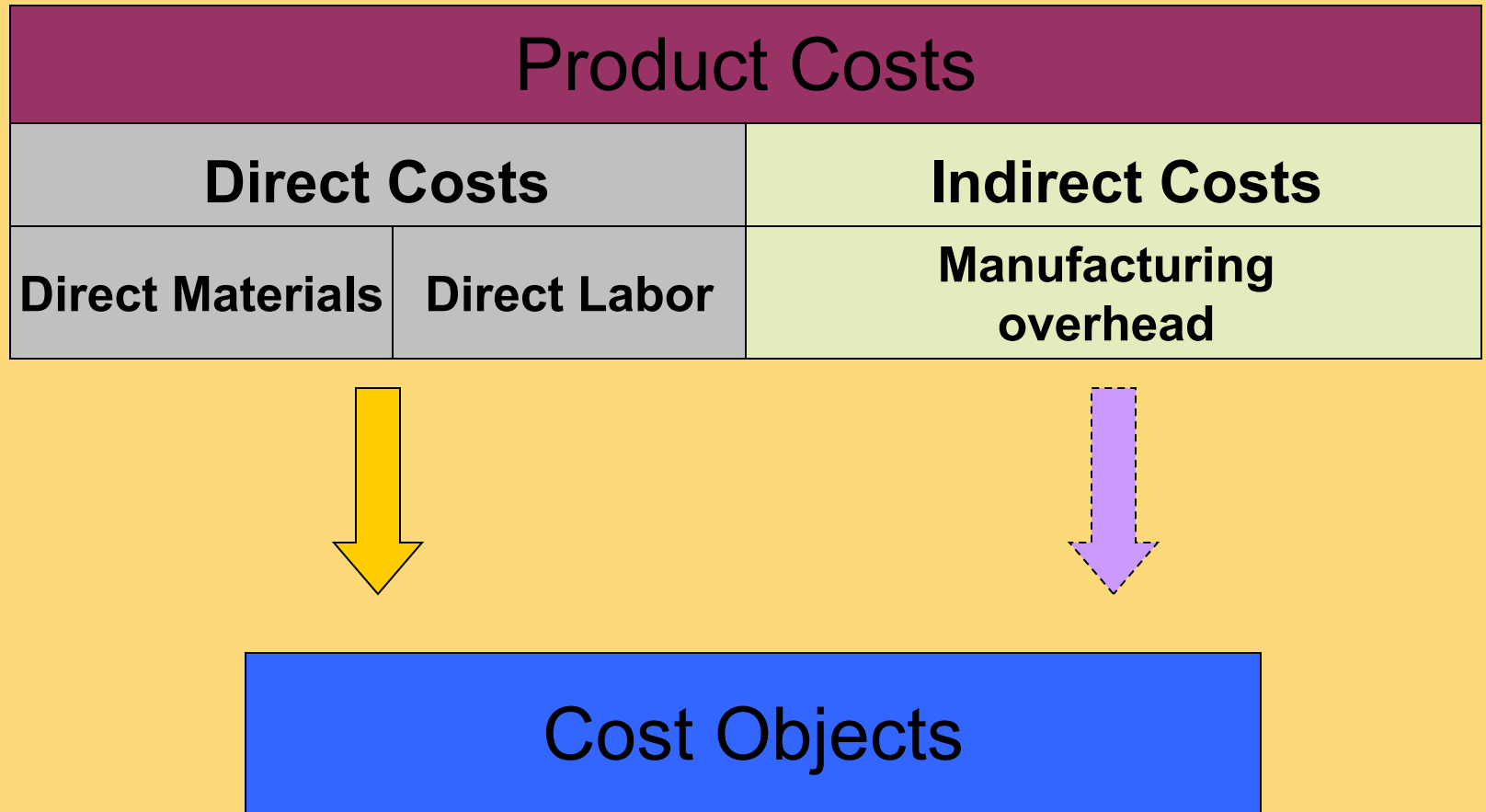
Chapter 8



Traditional Costing

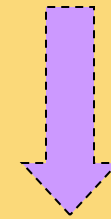
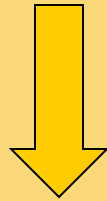


Traditional Costing



ABC

Product Costs			
Direct Costs		Indirect Costs	
Manufacturing	Non-manufacturing	Manufacturing	Non-manufacturing



Activities

Cost Objects

Activity Based Costing (ABC)

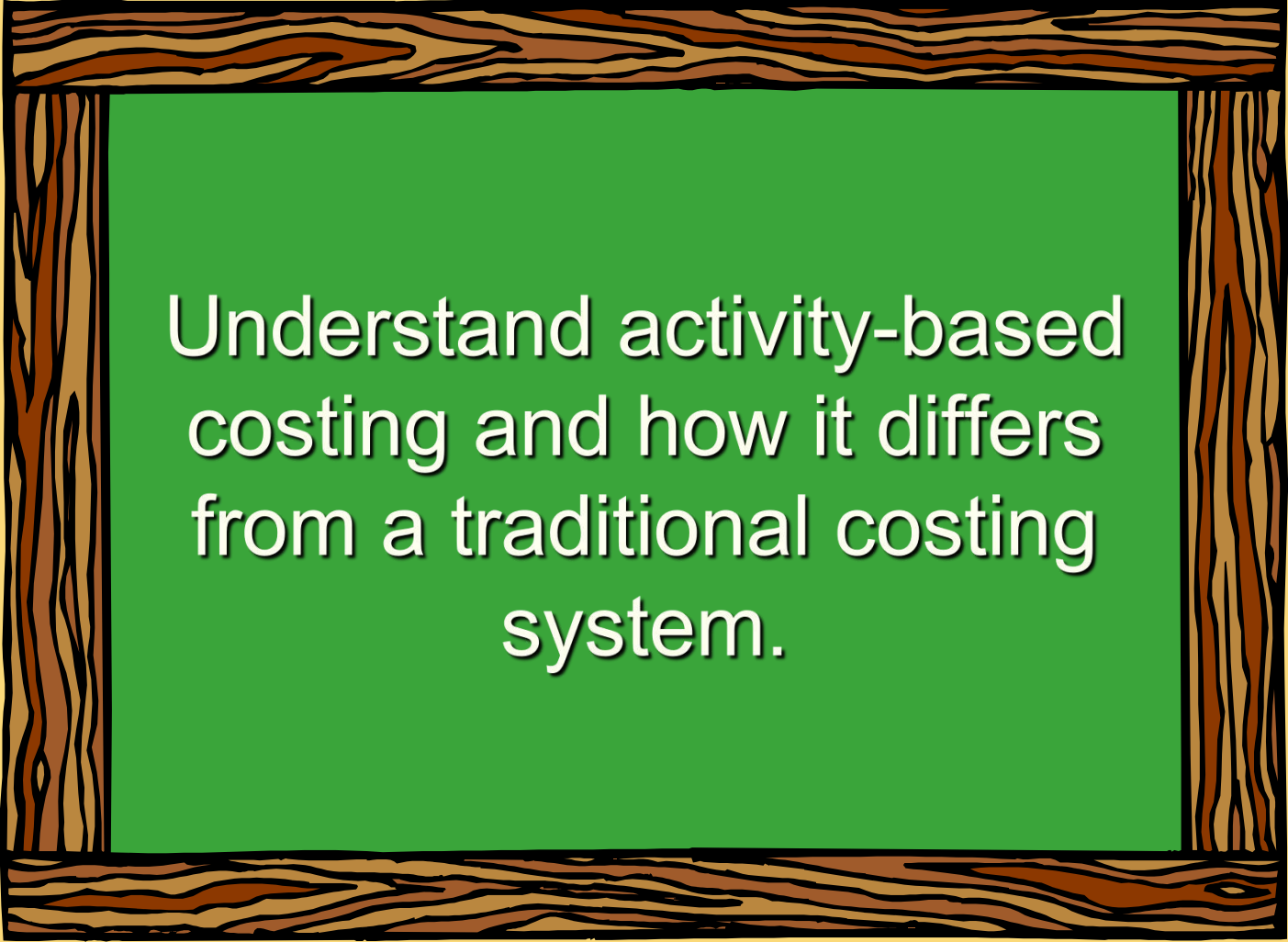
ABC is designed to provide managers with cost information for strategic and other decisions that potentially affect capacity and therefore affect fixed as well as variable costs.



Absorption, Variable Costing and ABC

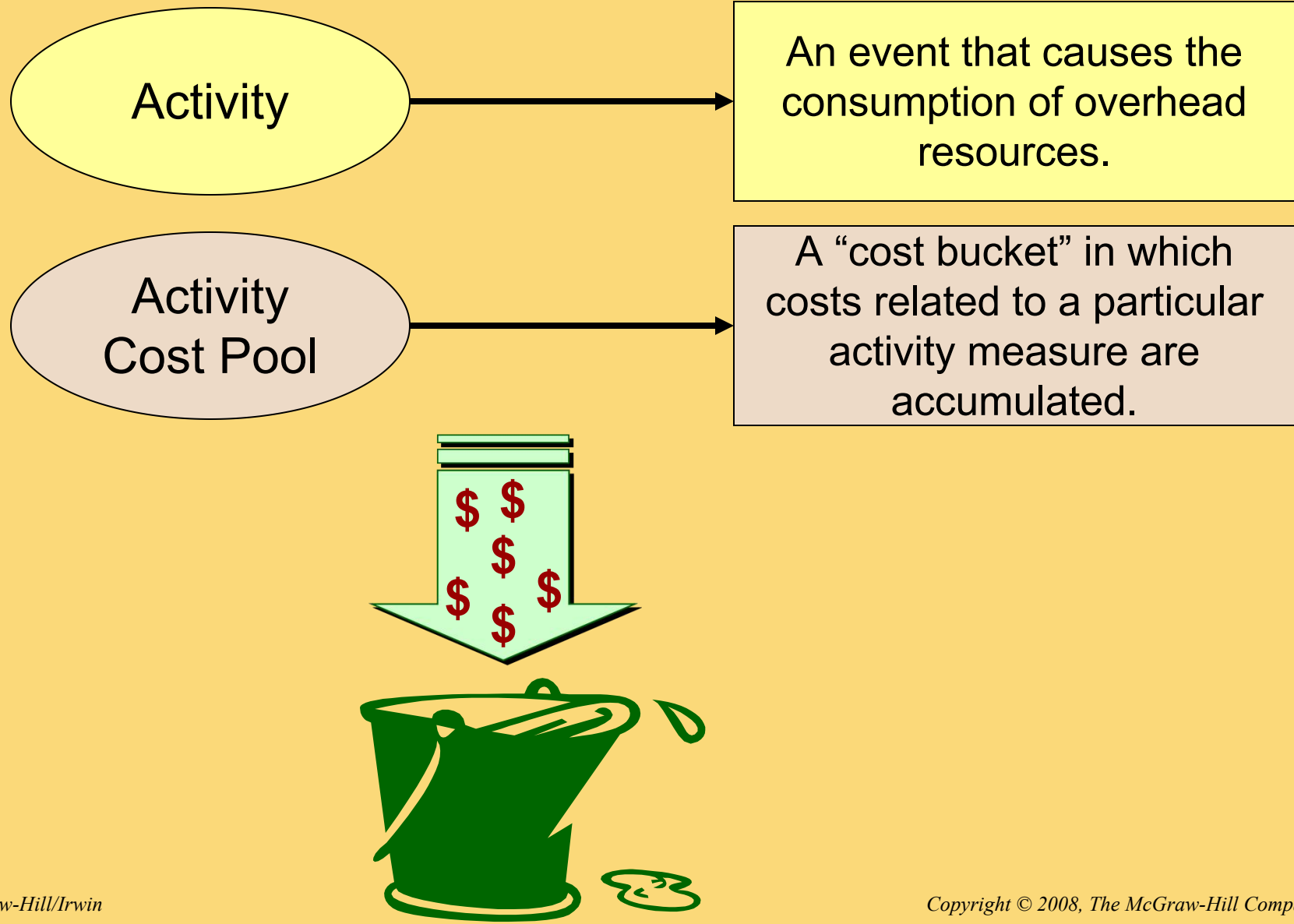
- Absorption costing is used for external reporting purposes
- Variable costing and ABC are used for internal decision making.
 - ◆ **Variable costing doesn't affect fixed costs**
 - ◆ **ABC affects fixed costs as well as variable costs.**

Learning Objective 1



Understand activity-based costing and how it differs from a traditional costing system.

How Costs are Treated Under Activity-Based Costing



How Costs are Treated Under Activity-Based Costing

Activity
Measure

The term cost driver is also used to refer to an activity measure.

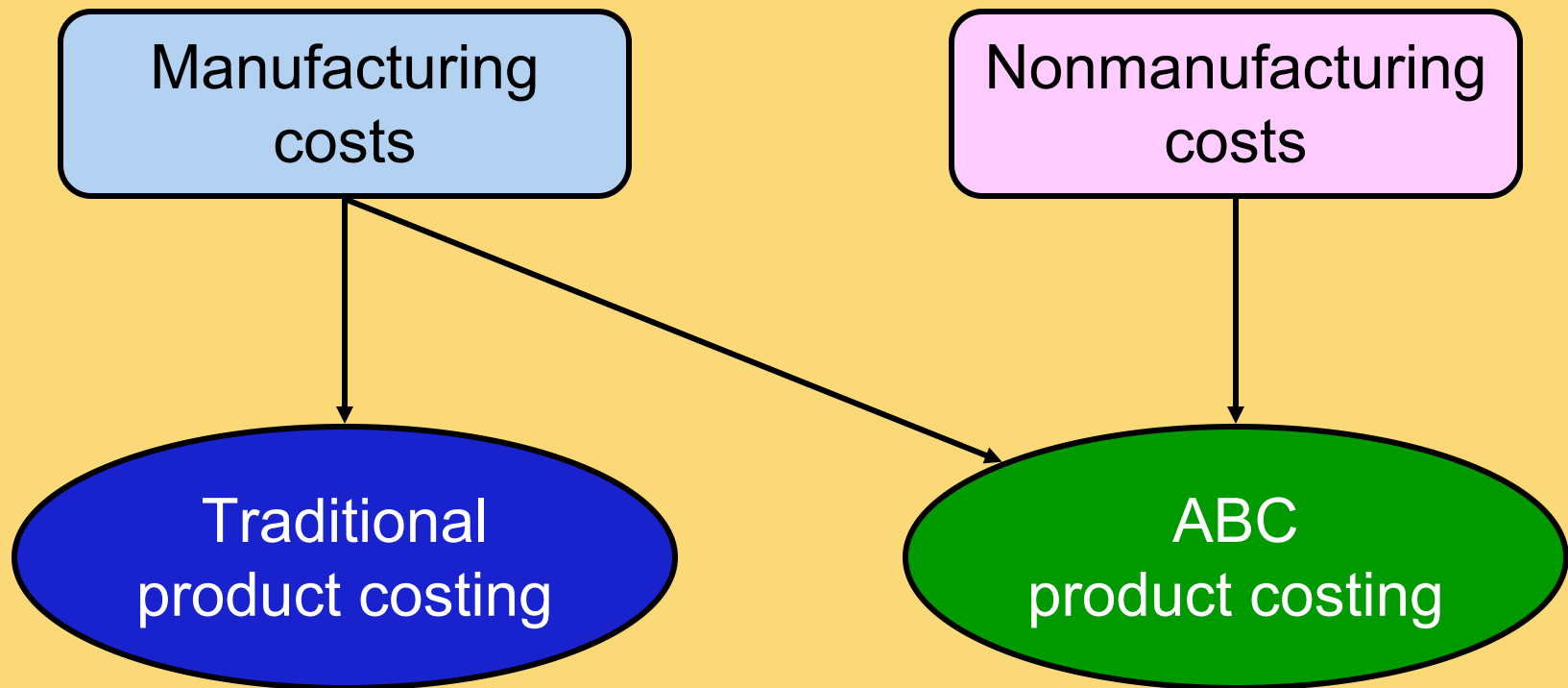
An allocation base
in an activity-based
costing system.



How Costs are Treated Under Activity-Based Costing

ABC differs from traditional cost accounting in three ways.

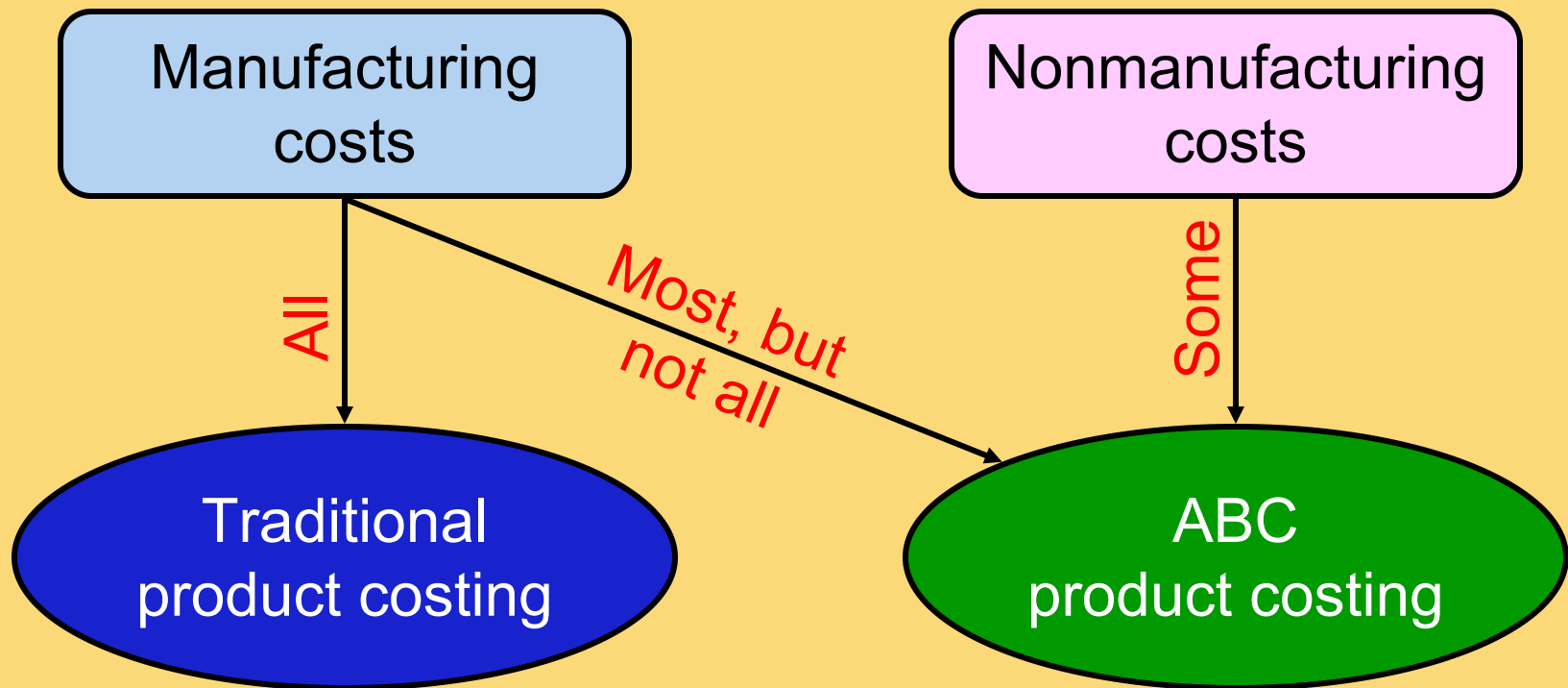
Overhead



❶ ABC assigns both types of costs to products.

How Costs are Treated Under Activity-Based Costing

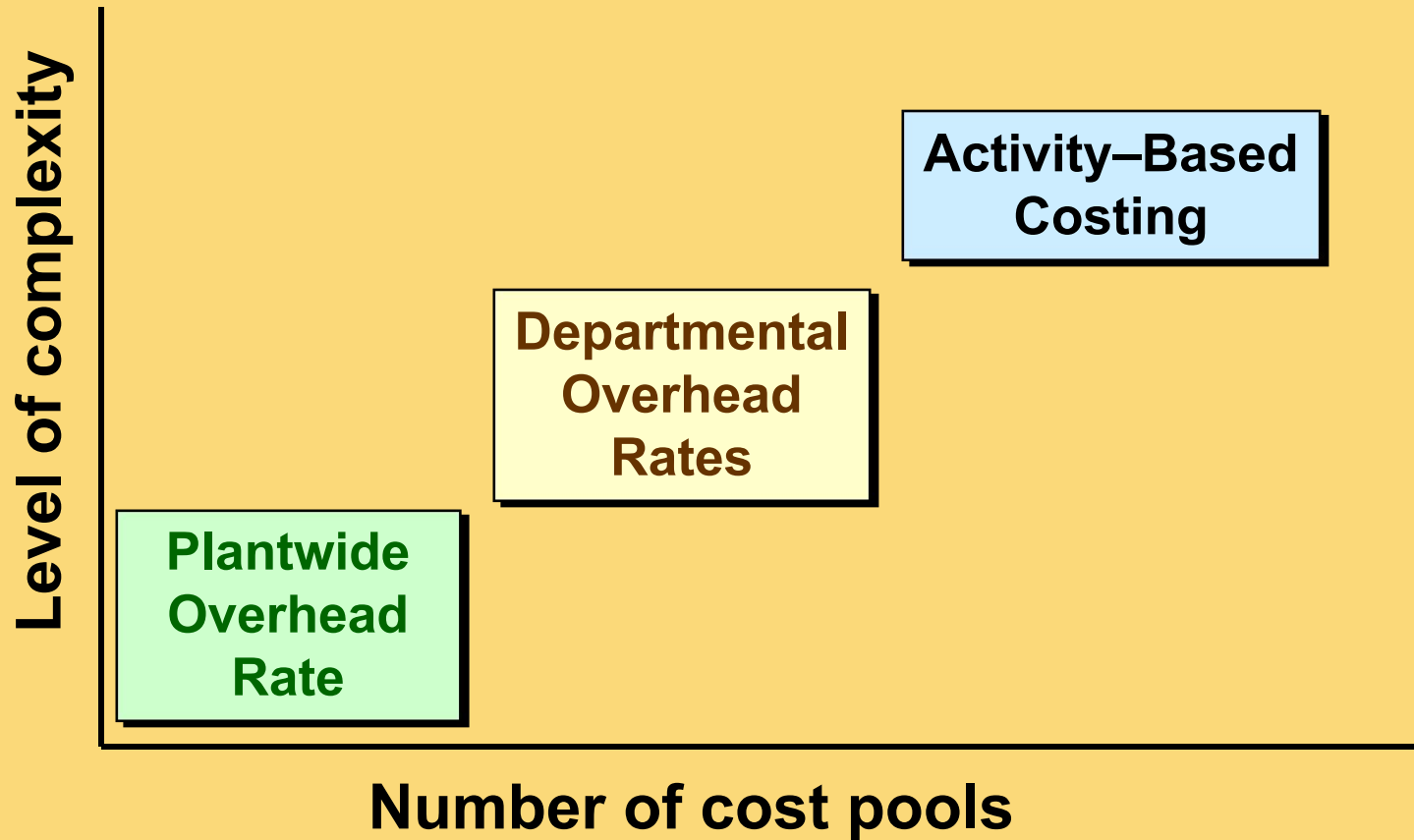
ABC differs from traditional cost accounting in three ways.



② ABC does not assign all manufacturing costs to products.

How Costs are Treated Under Activity-Based Costing

ABC differs from traditional cost accounting in three ways.



③ ABC uses more cost pools.

How Costs are Treated Under Activity-Based Costing

ABC differs from traditional cost accounting in three ways.

Each ABC cost pool has its own unique measure of activity.

Traditional cost systems usually rely on volume measures such as direct labor hours and/or machine hours to allocate all overhead costs to products.

③ ABC uses more cost pools.

How Costs are Treated Under Activity-Based Costing

ABC defines five levels of activity that largely do not relate to the volume of units produced.



Traditional cost systems usually rely on volume measures such as direct labor hours and/or machine hours to allocate all overhead costs to products.

How Costs are Treated Under Activity-Based Costing



A Case

- Western River runs river rafting trips. The company runs trips of one or two rafts, each of which carries two guides and up to 18 guests. The company provides all meals on the trip, which are prepared by the guides. Each guest is given a mug to use during the trip and to take home at the end of the trip as a souvenir. A guest can be considered as a unit and a raft as a batch.
 - ◆ the wages paid to the guides
 - ◆ the cost of a mug
 - ◆ The costs of food served to guests and guides

Characteristics of Successful ABC Implementations



**Strong top
management support**



**Link to evaluations
and rewards**



**Cross-functional
involvement**

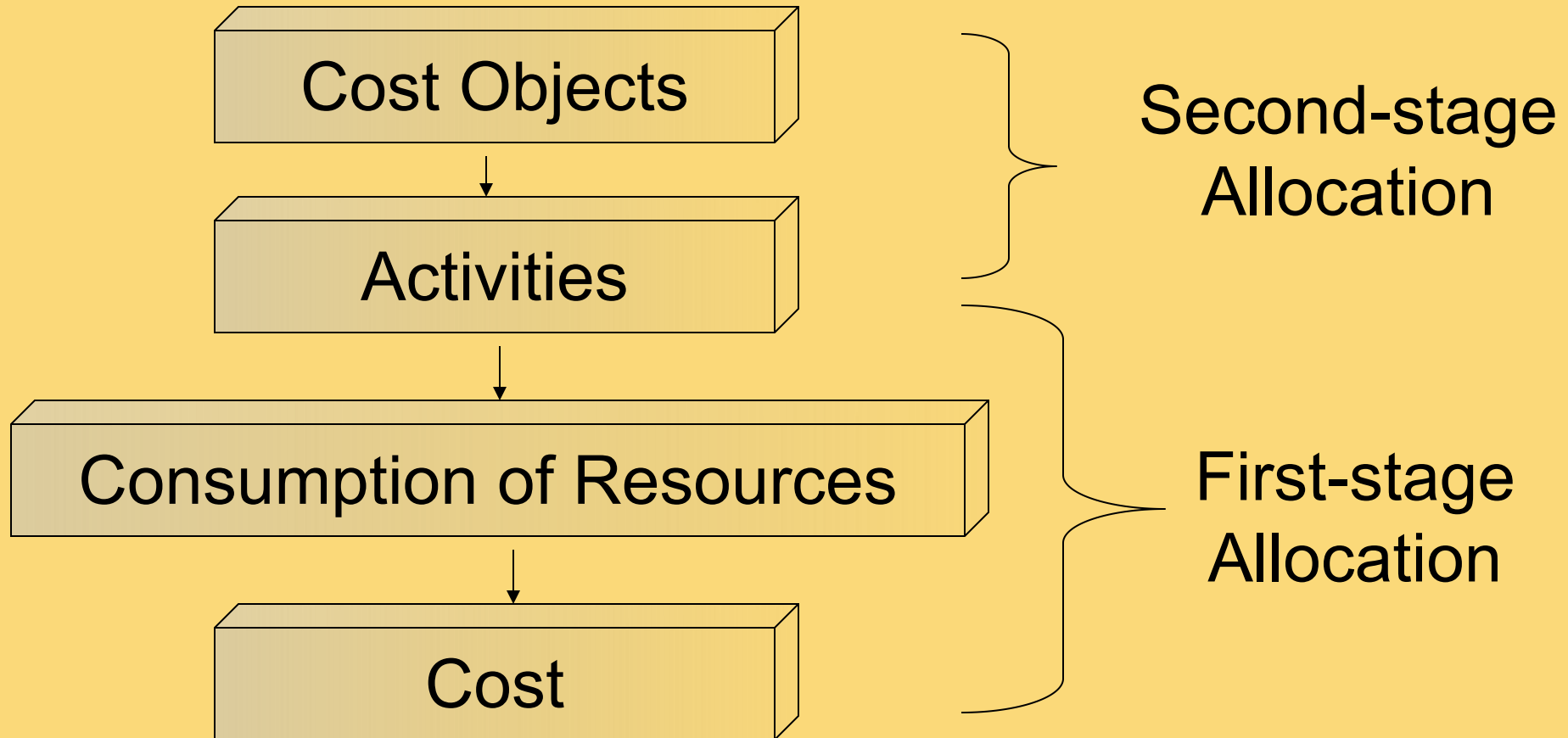
Classic Brass – An ABC Example

Classic Brass Income Statement Year Ended December 31, 2005

Sales		\$ 3,200,000
Cost of goods sold		
Direct materials	\$ 975,000	
Direct labor	351,250	
Manufacturing overhead	<u>1,000,000</u>	<u>2,326,250</u>
Gross margin		873,750
Selling and administrative expenses		
Shipping expenses	65,000	
Marketing expenses	300,000	
General administrative expenses	<u>510,000</u>	<u>875,000</u>
Net operating loss		<u><u>\$ (1,250)</u></u>

Manufacturing overhead is allocated to products using a single plantwide overhead rate based on machine hours.

ABC Model



① Define Activities, Activity Cost Pools, and Activity Measures

At Classic Brass, the ABC team, selected the following activity cost pools and activity measures:

Activity Cost Pools at Classic Brass

<u>Activity Cost Pool</u>		<u>Activity Measure</u>
Customer orders	—————▶	Number of customer orders
Product design	—————▶	Number of product designs
Order size	—————▶	Machine-hours
Customer relations	—————▶	Number of active customers
Other	—————▶	Not applicable

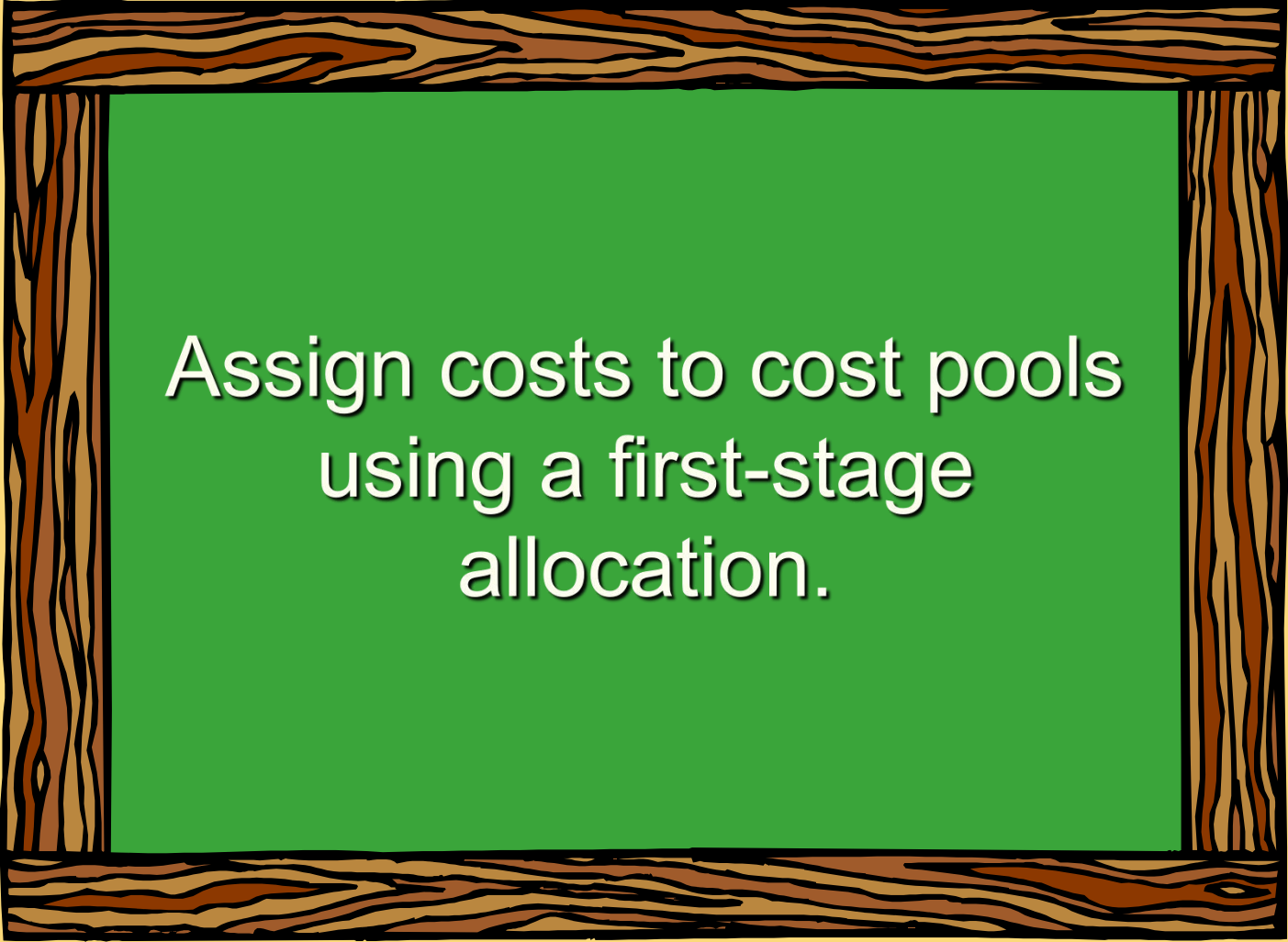


① Define Activities, Activity Cost Pools, and Activity Measures

- **Customer Orders** - assigned all costs of resources that are consumed by taking and processing customer orders.
- **Product Designs** - assigned all costs of resources consumed by designing products.
- **Order Size** - assigned all costs of resources consumed as a consequence of the number of units produced.
- **Customer Relations** – assigned all costs associated with maintaining relations with customers.
- **Other** – assigned all overhead costs that are not associated with the other cost pools.

Please tell the activity level for each of them.

Learning Objective 2



Assign costs to cost pools
using a first-stage
allocation.

② Assign Overhead Costs to Activity Cost Pools

Overhead Costs at Classic Brass (Manufacturing and Nonmanufacturing)

Production Department

Indirect factory wages	\$	500,000	
Factory equipment depreciation		300,000	
Factory utilities		120,000	
Factory building lease		80,000	\$ 1,000,000

General Administrative Department

Administrative wages and salaries		400,000	
Office equipment depreciation		50,000	
Administrative building lease		60,000	510,000

Marketing Department

Marketing wages and salaries		250,000	
Selling expenses		50,000	300,000

Total overhead costs			\$ 1,810,000
-----------------------------	--	--	---------------------

② Assign Overhead Costs to Activity Cost Pools

Overhead Costs at Classic Brass (Manufacturing and Nonmanufacturing)

Production Department

Indirect factory wages	\$	500,000	
Factory equipment depreciation		300,000	
Factory utilities		120,000	
Factory building lease		80,000	\$ 1,000,000

General Administrative Department

Administrative wages and salaries		400,000	
Office equipment depreciation		50,000	
Administrative building lease		60,000	510,000

Marketing Department

Direct materials, direct labor, and shipping are excluded because Classic Brass' existing cost system can directly trace these costs to products or customer orders.

② Assign Overhead Costs to Activity Cost Pools

At Classic Brass the following distribution of resource consumption across activity cost pools is determined.

Activity Cost Pools						
	Customer Orders	Product Design	Order Size	Customer Relations	Other	Total
Production Department						
Indirect factory wages	25%	40%	20%	10%	5%	100%
Factory equipment depreciation	20%	0%	60%	0%	20%	100%
Factory utilities	0%	10%	50%	0%	40%	100%
Factory building lease	0%	0%	0%	0%	100%	100%
General Administrative Department						
Administrative wages and salaries	15%	5%	10%	30%	40%	100%
Office equipment depreciation	30%	0%	0%	25%	45%	100%
Administrative building lease	0%	0%	0%	0%	100%	100%
Marketing Department						
Marketing wages and salaries	22%	8%	0%	60%	10%	100%
Selling expenses	10%	0%	0%	70%	20%	100%

② Assign Overhead Costs to Activity Cost Pools

	Activity Cost Pool
	Customer Orders
Production Department	
Indirect factory wages	\$ 125,000
Factory equipment depreciation	
Factory utilities	
Factory building lease	
General Administrative Department	
Administrative wages and salaries	
Office equipment depreciation	
Administrative building lease	
Marketing Department	
Marketing wages and salaries	
Selling expenses	
Total	

Overhead Costs at Classic Brass (Manufacturing and Nonmanufacturing)

Production Department			
Indirect factory wages	\$	500,000	
Factory equipment depreciation		300,000	
Factory utilities		120,000	
Factory building lease		80,000	\$ 1,000,000
General Administrative Department			
Administrative wages and salaries		400,000	
Office equipment depreciation		50,000	
Administrative building lease		60,000	510,000
Marketing Department			
Marketing wages and salaries		250,000	
Selling expenses		50,000	300,000
Total overhead costs			<u>\$ 1,810,000</u>

Indirect factory wages	\$500,000
Percent consumed by customer orders	25%
	<u>\$125,000</u>

② Assign Overhead Costs to Activity Cost Pools

	Activity Cost Pool
	Customer Orders
Production Department	
Indirect factory wages	\$ 125,000
Factory equipment depreciation	60,000
Factory utilities	
Factory building lease	
General Administrative Department	
Administrative wages and salaries	
Office equipment depreciation	
Administrative building lease	
Marketing Department	
Marketing wages and salaries	
Selling expenses	
Total	

Overhead Costs at Classic Brass (Manufacturing and Nonmanufacturing)

Production Department		
Indirect factory wages	\$ 500,000	
Factory equipment depreciation	300,000	
Factory utilities	120,000	
Factory building lease	80,000	\$ 1,000,000
General Administrative Department		
Administrative wages and salaries	400,000	
Office equipment depreciation	50,000	
Administrative building lease	60,000	510,000
Marketing Department		
Marketing wages and salaries	250,000	
Selling expenses	50,000	300,000
Total overhead costs		<u>\$ 1,810,000</u>

Factory equipment depreciation	\$300,000
Percent consumed by customer orders	20%
	<u>\$ 60,000</u>

② Assign Overhead Costs to Activity Cost Pools

	Activity Cost Pools					Total
	Customer Orders	Product Design	Order Size	Customer Relations	Other	
Production Department						
Indirect factory wages	\$ 125,000	\$ 200,000	\$ 100,000	\$ 50,000	\$ 25,000	\$ 500,000
Factory equipment depreciation	60,000	-	180,000	-	60,000	300,000
Factory utilities	-	12,000	60,000	-	48,000	120,000
Factory building lease	-	-	-	-	80,000	80,000
General Administrative Department						
Administrative wages and salaries	60,000	20,000	40,000	120,000	160,000	400,000
Office equipment depreciation	15,000	-	-	12,500	22,500	50,000
Administrative building lease	-	-	-	-	60,000	60,000
Marketing Department						
Marketing wages and salaries	55,000	20,000	-	150,000	25,000	250,000
Selling expenses	5,000	-	-	35,000	10,000	50,000
Total	\$ 320,000	\$ 252,000	\$ 380,000	\$ 367,500	\$ 490,500	\$ 1,810,000

Learning Objective 3

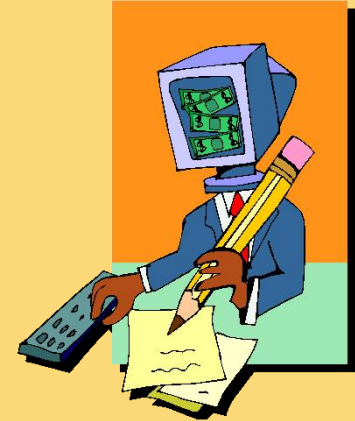


Compute activity
rates for cost pools.

③ Calculate Activity Rates

The ABC team determines that Classic Brass will have these total activities for each activity cost pool . . .

- ◆ 1,000 customer orders,
- ◆ 400 new designs,
- ◆ 20,000 machine-hours,
- ◆ 250 customer relations activities.



Now the team can compute the individual activity rates by dividing the total cost for each activity by the total activity levels.

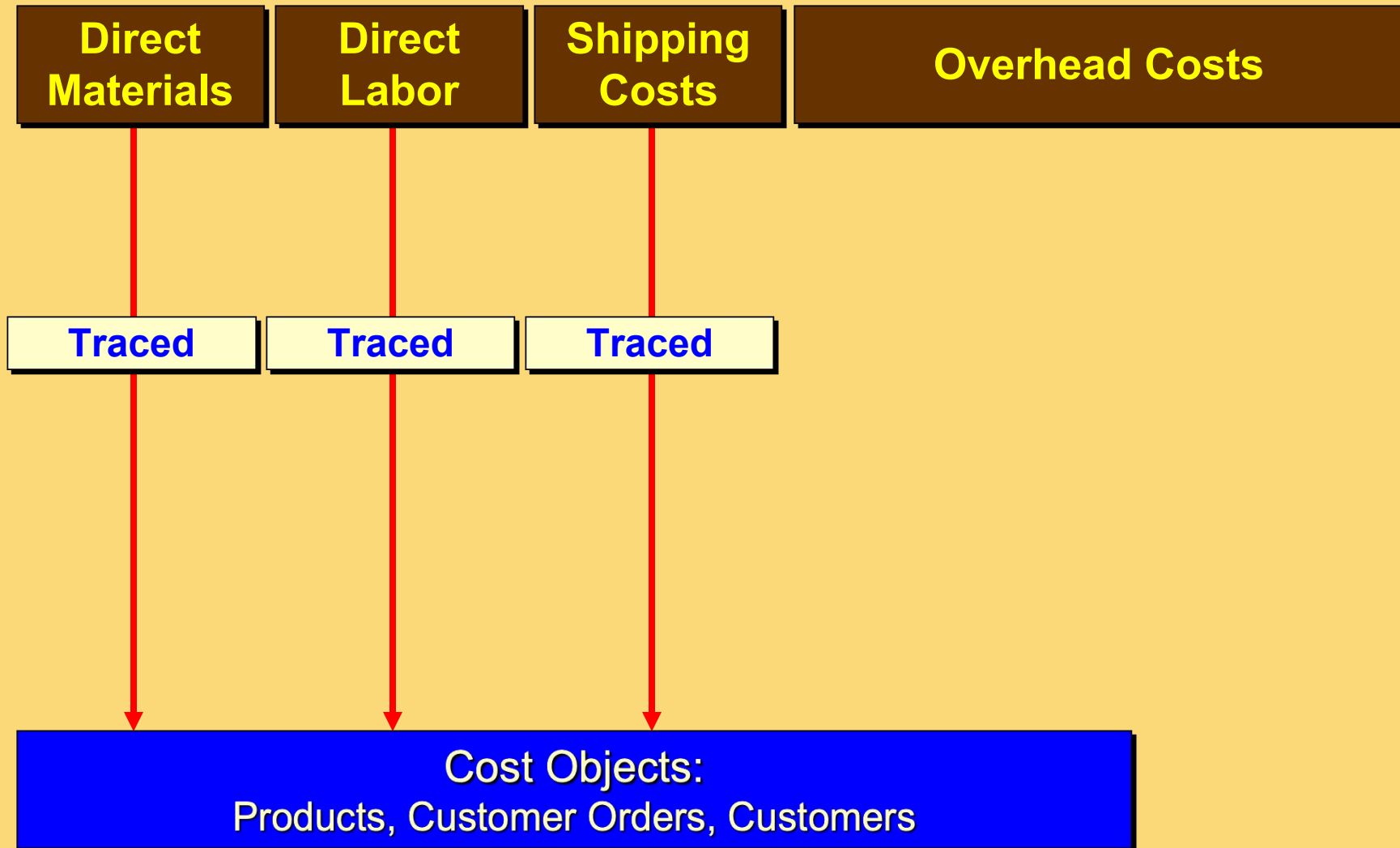
③ Calculate Activity Rates

Computation of Activity Rates

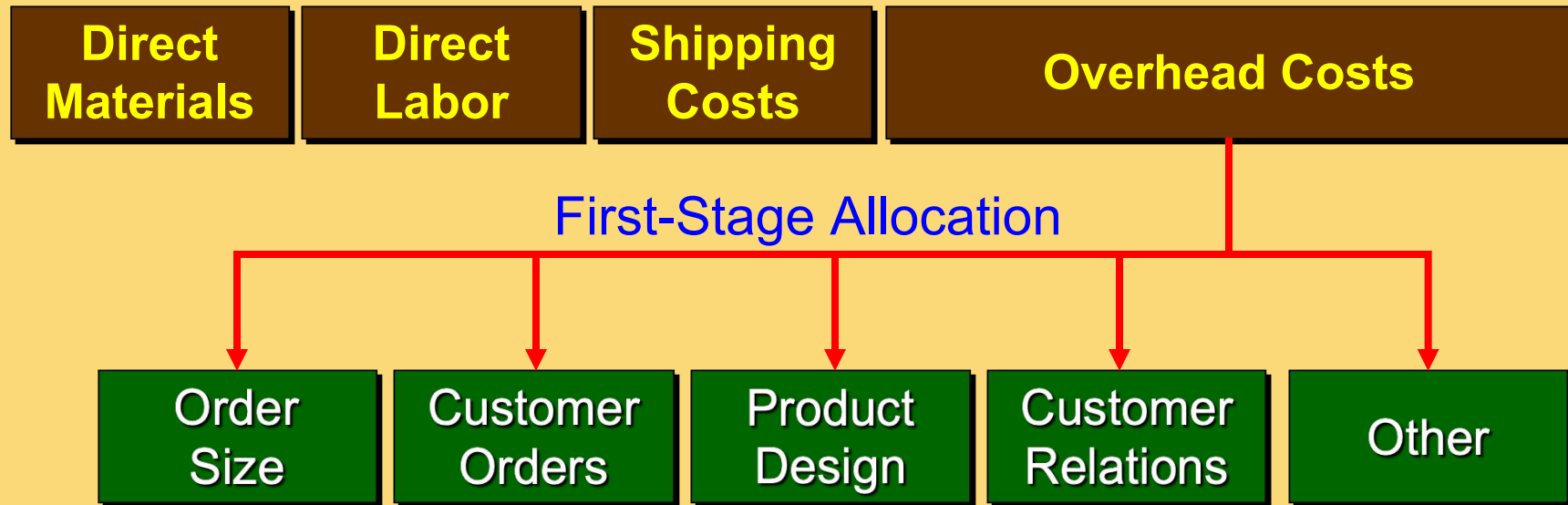
<i>Activity Cost Pools</i>	(a) <i>Total Cost</i>	(b) <i>Total Activity</i>	(a) ÷ (b) <i>Activity Rate</i>
Customer orders	\$ 320,000	1,000 orders	\$320 per order
Product design	252,000	400 designs	\$630 per design
Order size	380,000	20,000 MHs	\$19 per MH
Customer relations	367,500	250 customer	\$1,470 per customer
Other	490,500	Not applicable	Not applicable



Activity-Based Costing at Classic Brass

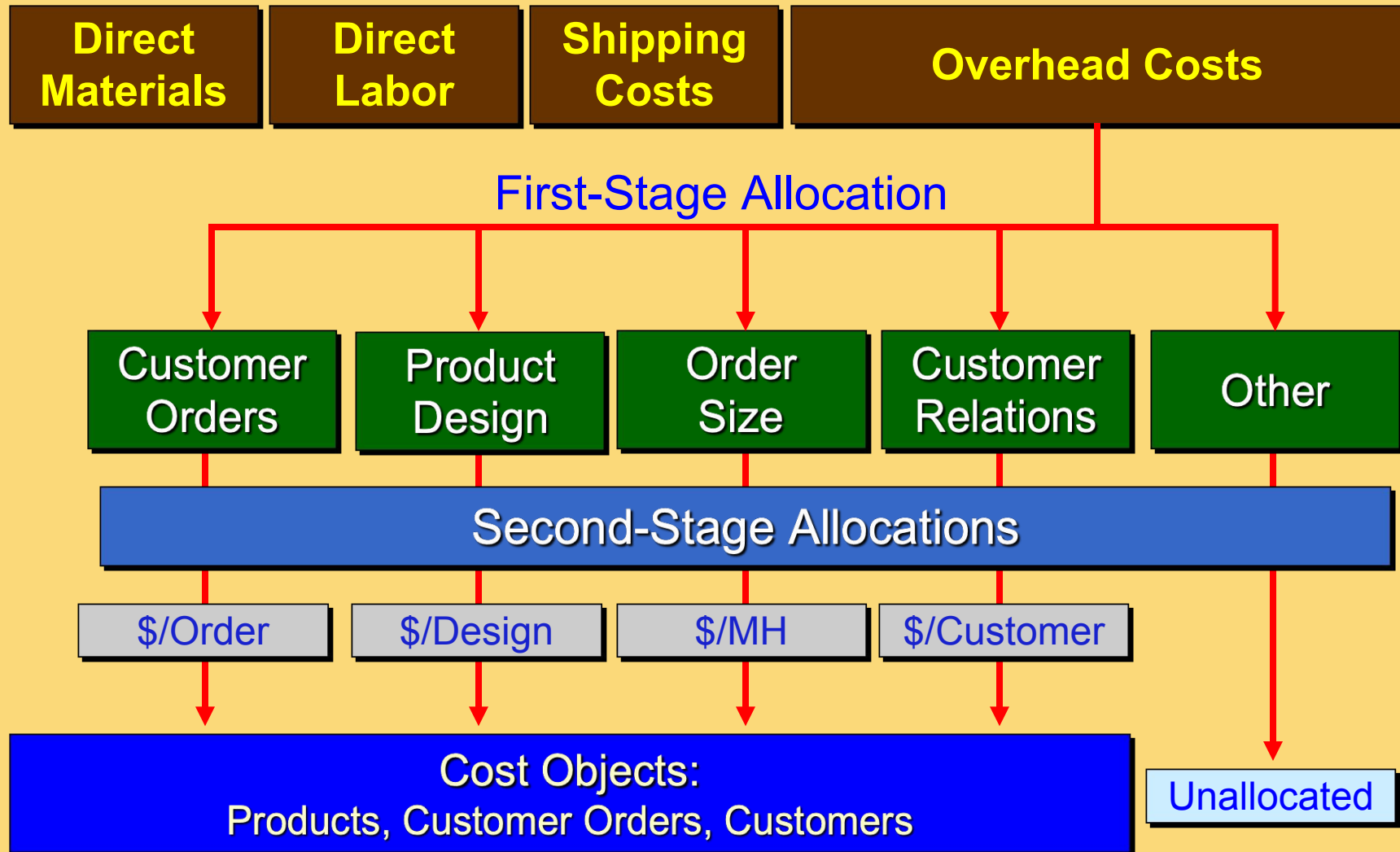


Activity-Based Costing at Classic Brass

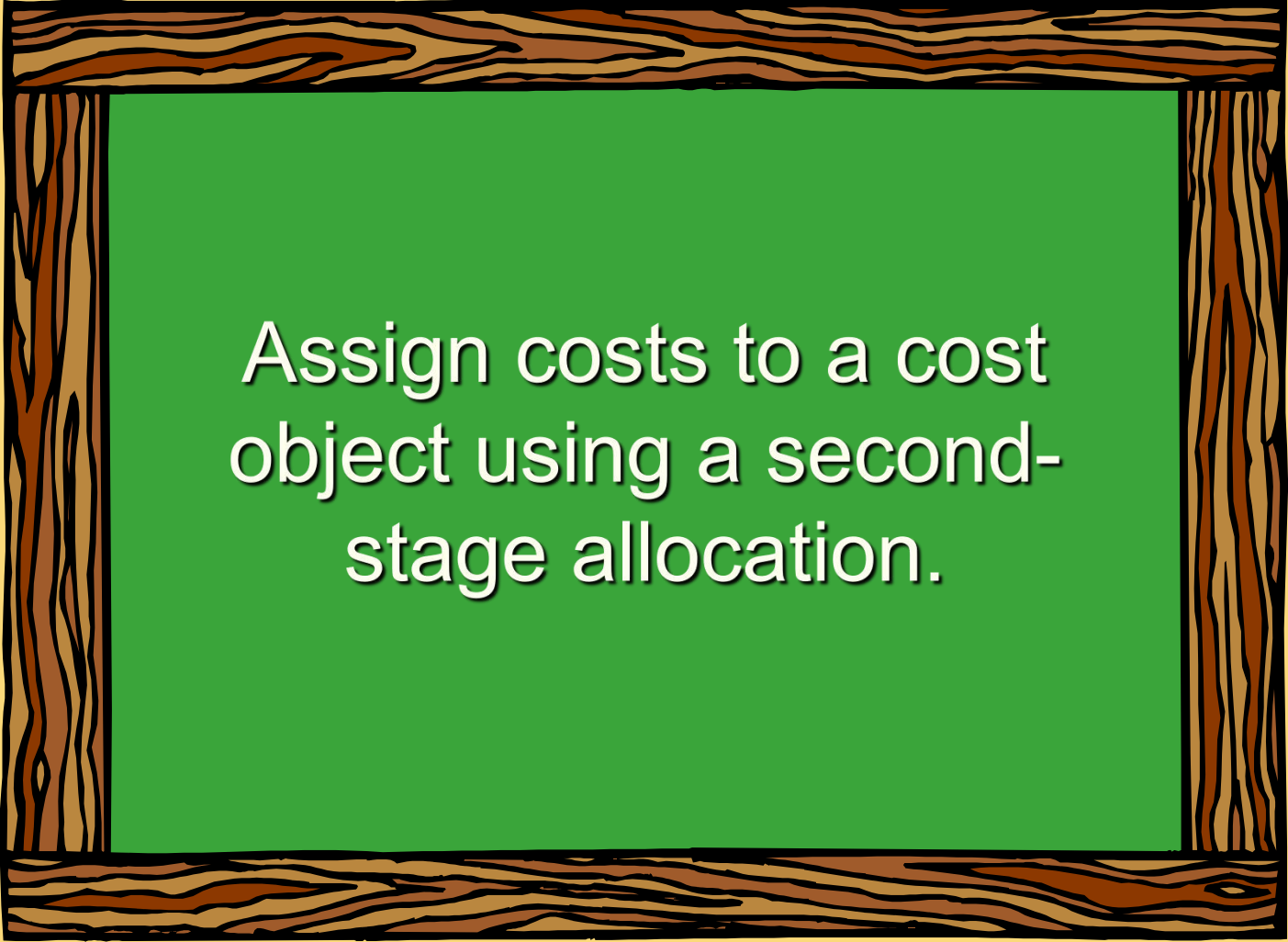


Cost Objects:
Products, Customer Orders, Customers

Activity-Based Costing at Classic Brass



Learning Objective 4



Assign costs to a cost object using a second-stage allocation.

④ Assigning Overhead to Products

Classic Brass Information

Standard Stanchions

1. Requires no new design resources.
2. 30,000 units ordered with 600 separate orders.
3. Each stanchion requires 35 minutes of machine time for a total of 17,500 machine-hours.

Custom Compass Housing

1. Requires new design resources.
2. 400 separate orders.
3. 400 product designs prepared.
4. 1,250 compass housings produced, requiring 2 machine-hours each for a total of 2,500 machine-hours.

④ Assigning Overhead to Products

Overhead Cost for the Standard Stanchions

<i>Activity Cost Pools</i>	(a) <i>Activity Rate</i>	(b) <i>Activity</i>	(a) × (b) <i>ABC Cost</i>
Customer orders	\$ 320	600	\$ 192,000
Product design	630	0	-
Order size	19	17500	332,500
Total			\$ 524,500

Overhead Cost for the Custom Housing

<i>Activity Cost Pools</i>	(a) <i>Activity Rate</i>	(b) <i>Activity</i>	(a) × (b) <i>ABC Cost</i>
Customer orders	\$ 320	400	\$ 128,000
Product design	630	400	252,000
Order size	19	2500	47,500
Total			\$ 427,500

Total Overhead Cost			
Total Overhead Assigned	Standard Stanchions	\$524,500	\$952,000
	Custom Compass Housing	\$427,500	
Total Overhead not Assigned	Custum Relations Activity	\$367,500	\$858,000
	Other Activity	\$490,500	
Total			\$1,810,000

Shown
in Slide
18 & 23

Assigning Overhead to Customers

Let's take a look at how Classic Brass system works for just one of the 250 customers – Windward Yachts who placed a total of three orders.

Orders

1. Two orders for 150 standard stanchions per order.
2. One order for a custom compass housing.

Machine-hours

1. The 300 standard stanchions required 175 machine-hours.
2. The custom compass housing required 2 machine hours.

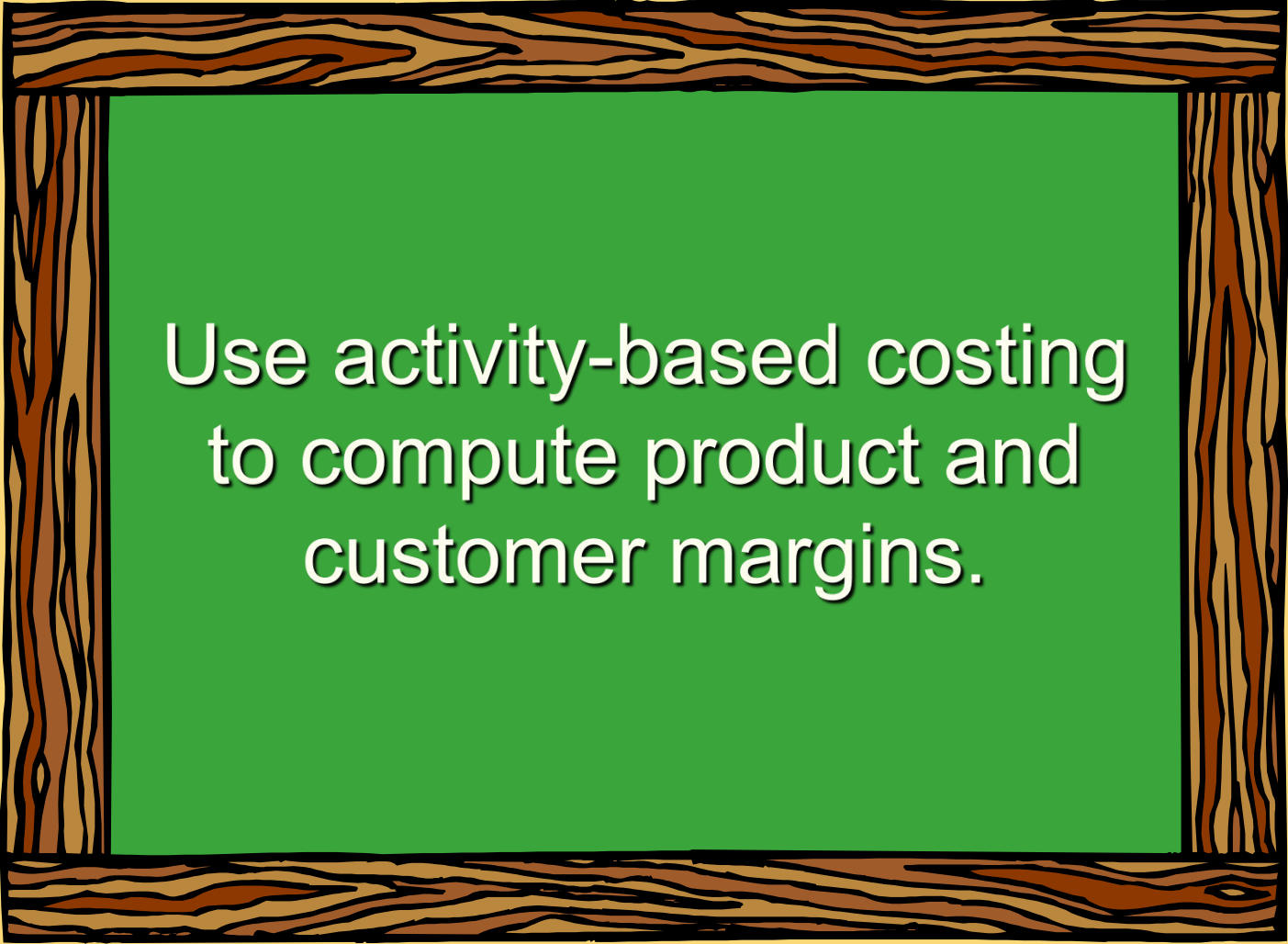
Assigning Overhead to Customers

Overhead Cost for Winward Yachts

<i>Activity Cost Pools</i>	(a) <i>Activity Rate</i>	(b) <i>Activity</i>	(a) × (b) <i>ABC Cost</i>
Customer orders	\$ 320	3	\$ 960
Product design	630	1	630
Order size	19	177	3,363
Customer relations	1,470	1	1,470
Total			\$ 6,423



Learning Objective 5



Use activity-based costing
to compute product and
customer margins.

⑤ Prepare Management Reports

Product Margin Calculations

The first step in computing product margins is to gather each product's sales and direct cost data.

	Standard Stanchions	Custom Compass Housings	Total
Sales	\$ 2,660,000	\$ 540,000	\$ 3,200,000
Direct costs			
Direct material	905,500	69,500	975,000
Direct labor	263,750	87,500	351,250
Shipping	60,000	5,000	65,000

⑤ Prepare Management Reports

Product Margin Calculations

The second step in computing product margins is to incorporate the previously computed **activity-based cost assignments** pertaining to each product.

	Standard Stanchions	Custom Compass Housings	Total
Sales	\$ 2,660,000	\$ 540,000	\$ 3,200,000
Direct costs			
Direct material	905,500	69,500	975,000
Direct labor	263,750	87,500	351,250
Shipping	60,000	5,000	65,000
ABC cost assignments			
Customer orders	192,000	128,000	320,000
Product design		252,000	252,000
Order size	332,500	47,500	380,000

⑤ Prepare Management Reports

Product Margin Calculations

The third step in computing product margins is to deduct each product's direct and indirect costs from sales.

	Standard Stanchions	Custom Compass Housings
Sales	\$ 2,660,000	\$ 540,000
Costs		
Direct material	\$ 905,500	\$ 69,500
Direct labor	263,750	87,500
Shipping	60,000	5,000
Customer orders	192,000	128,000
Product design		252,000
Order size	332,500	47,500
Total cost	1,753,750	589,500
Product margin	\$ 906,250	(49,500)

⑤ Prepare Management Reports

Product Margin Calculations

The product margins can be reconciled with the company's net operating income as follows:

	Standard Stanchions	Custom Compass Housings	Total
Sales	\$ 2,660,000	\$ 540,000	\$ 3,200,000
Total costs	1,753,750	589,500	2,343,250
Product margins	<u>\$ 906,250</u>	<u>\$ (49,500)</u>	<u>\$ 856,750</u>
Less costs not assigned to products:			
Customer relations			367,500
Other			490,500
Total			<u>858,000</u>
Net operating loss			<u>\$ (1,250)</u>

⑤ Prepare Management Reports

Customer Profitability Analysis

The first step in computing Windward Yachts' customer margin is to gather its sales and direct cost data.

	Windward Yachts
Sales	\$ 11,350
Direct costs	
Direct material	2,123
Direct labor	1,900
Shipping	205

⑤ Prepare Management Reports

Customer Profitability Analysis

The second step is to incorporate Windward Yachts' previously computed **activity-based cost assignments**.

	Windward Yachts
Sales	\$ 11,350
Direct costs	
Direct material	2,123
Direct labor	1,900
Shipping	205
ABC cost assignments	
Customer orders	960
Product design	630
Order size	3,363
Customer relations	1,470

⑤ Prepare Management Reports

Customer Profitability Analysis

The third step is to compute Windward Yachts' customer margin (\$699) by deducting all its direct and indirect costs from its sales.

	<u>Windward Yachts</u>	
Sales		\$ 11,350
Direct costs		
Direct material	\$ 2,123	
Direct labor	1,900	
Shipping	205	
Customer orders	960	
Product design	630	
Order size	3,363	
Customer relations	1,470	10,651
Customer margin		\$ 699

A Case

- Finding that Golden top 20%
 - ◆ 20% of a bank's customers generate about 150% of its profits; At the other end of the spectrum, 30% of a bank's customers drain 50% of its profits.

Product Margins Computed Using the Traditional Cost System

The first step in computing product margins is to gather each product's sales and direct cost data.

	Standard Stanchions	Custom Compass Housings	Total
Sales	\$ 2,660,000	\$ 540,000	\$ 3,200,000
Direct costs			
Direct material	905,500	69,500	975,000
Direct labor	263,750	87,500	351,250



Product Margins Computed Using the Traditional Cost System

The second step in computing product margins is to compute the plantwide overhead rate.

Manufacturing Overhead Costs at Classic Brass

Production Department

Indirect factory wages	\$ 500,000
Factory equipment depreciation	300,000
Factory utilities	120,000
Factory building lease	80,000
Total manufacturing overhead	<u><u>\$ 1,000,000</u></u>

$$\text{Plantwide manufacturing overhead rate} = \frac{\$1,000,000}{20,000 \text{ MH}} = \$50 \text{ per machine-hour}$$

Standard Stanchions
Custom compass Housings
Total machine-hours

Machine-hours
17,500
2,500
<u><u>20,000</u></u>

Product Margins Computed Using the Traditional Cost System

The third step in computing product margins is allocate manufacturing overhead to each product.

	Machine Hours	Overhead Rate	Overhead Allocated
Standard Stanchions	17,500	\$ 50.00	\$ 875,000
Custom Compass Housings	2,500	50.00	125,000
Total overhead allocated to products			<u>\$ 1,000,000</u>

$$17,500 \text{ hours} \times \$50 \text{ per hour} = \$875,000$$



Product Margins Computed Using the Traditional Cost System

The fourth step is to actually compute the product margins.

	Standard Stanchions	Custom Compass Housings	Total
Sales	\$ 2,660,000	\$ 540,000	\$ 3,200,000
Cost of goods sold			
Direct materials	\$ 905,500	\$ 69,500	\$ 975,000
Direct labor	263,750	87,500	351,250
Manufacturing overhead	875,000	125,000	1,000,000
	<u>2,044,250</u>	<u>282,000</u>	<u>2,326,250</u>
Product margin	<u>\$ 615,750</u>	<u>258,000</u>	873,750
Selling and administrative			875,000
Net operating loss			<u>\$ (1,250)</u>

Shipping expenses	\$ 65,000
Marketing expenses	300,000
General administrative expenses	510,000
	<u>\$ 875,000</u>

The Differences Between ABC and Traditional Product Costs

	Standard Stanchions	Custom Compass Housings
Product margins – traditional	\$ 615,750	\$ 258,000
Product margins – ABC	906,250	(49,500)
Change in reported margins	<u>\$ 290,500</u>	<u>\$ (307,500)</u>

The traditional cost system **overcosts** the standard stanchions and reports a **lower** product margin for this product.

The traditional cost system **undercosts** the custom compass housings and reports a **higher** product margin for this product.

Differences Between ABC and Traditional Product Costs

There are three reasons why the reported product margins for the two costing systems differ from one another.

- ❶ Traditional costing allocates all manufacturing overhead to products. ABC costing only assigns manufacturing overhead costs consumed by products to those products.

Differences Between ABC and Traditional Product Costs

There are three reasons why the reported product margins for the two costing systems differ from one another.

- ② Traditional costing allocates all manufacturing overhead costs using a volume-related allocation base. ABC costing also uses non-volume related allocation bases.

Differences Between ABC and Traditional Product Costs

There are three reasons why the reported product margins for the two costing systems differ from one another.

- ③ Traditional costing disregards selling and administrative expenses because they are assumed to be period expenses. ABC costing directly traces shipping costs to products and includes nonmanufacturing overhead costs caused by products in the activity cost pools that are assigned to products.

Targeting Process Improvement

Activity-based management is used in conjunction with ABC to identify areas that would benefit from process improvements.



While the **theory of constraints** approach discussed in Chapter 1 is a powerful tool for targeting improvement efforts, activity rates can also provide valuable clues on where to focus improvement efforts.

Benchmarking can be used to compare activity cost information with world-class standards of performance achieved by other organizations.

Activity-Based Costing and External Reporting

Most companies do not use ABC for external reporting because . . .

1. External reports are less detailed than internal reports.
2. It may be difficult to make changes to the company's accounting system.
3. ABC does not conform to GAAP.
4. Auditors may be suspect of the subjective allocation process based on interviews with employees.

ABC Limitations

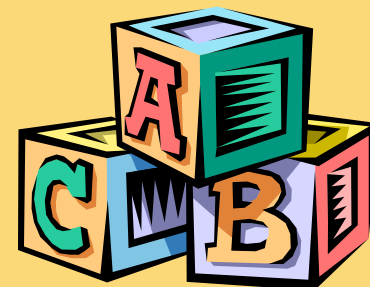
Substantial resources required to implement and maintain.

Resistance to unfamiliar numbers and reports.

Desire to fully allocate all costs to products.

Potential misinterpretation of unfamiliar numbers.

Does not conform to GAAP. Two costing systems may be needed.



- Implementing an ABC system requires substantial resources. The benefits of increased cost accuracy may not outweigh the implementation costs.
- ABC systems produce numbers, such as product margins, that are at odds with the numbers produced by traditional cost systems. Managers are not accustomed to managing their operations using these numbers; hence, ABC inevitably faces resistance. This underscores the importance of having top management support for and cross-functional involvement with the ABC implementation.
- In practice, most managers insist on fully allocating all costs to products. The ABC system described in the main portion of this chapter does not conform to this preference.

- ABC systems do not automatically identify the relevant costs for particular decisions; therefore, ABC data can be easily misinterpreted and must be used with care when making decisions. Costs assigned to products, customers, and other cost objects are only potentially relevant.
- Most organizations use ABC as a supplement to rather a replacement for their existing cost system. Maintaining two cost systems is more costly than maintaining just one system and it may cause confusion about which set of numbers is to be relied on.

A Problem

- Gore is a small, family-owned business. For its services, the company has always charged a flat fee per hundred square feet of carpet cleaned. The current fee is \$22.95 per hundred square feet. However, there is some question about whether the company is actually making any money on jobs for some customers—particularly those located on more remote ranches that require considerable travel time. The owner's daughter, home for the summer from college, has suggested investigating this question using activity-based costing. After some discussion, a simple system consisting of four activity cost pools seemed to be adequate. The activity cost pools and their activity measures appear below:**

Activity Cost Pool	Activity Measure	Activity for the Year
Cleaning carpets	Square feet cleaned(00s)	10,000hundred square feet
Travel to jobs	Miles driven	50,000 miles
Job support	Number of jobs	1,800 jobs
Other (costs of idle capacity and organization-sustaining costs)	None	Not applicable

- The total cost of operating the company for the year is \$340,000, which includes the following costs:

Wages	\$140,000
Cleaning supplies	25,000
Cleaning equipment depreciation	10,000
Vehicle expenses	30,000
Office expenses	60,000
President's compensation	75,000
Total cost	\$340,000

- Resource consumption is distributed across the activities as follows:

	Distribution of Resource Consumption				
	Across Activity Cost Pools				
	Cleaning Carpets	Travel to Jobs	Job Support	Other	Total
Wages	75%	15%	0%	10%	100%
Cleaning supplies	100%	0%	0%	0%	100%
Cleaning equipment depreciation	70%	0%	0%	30%	100%
Vehicle expenses	0%	80%	0%	20%	100%
Office expenses	0%	0%	60%	40%	100%
President's compensation	0%	0%	30%	70%	100%

- **Job support consists of receiving calls from potential customers at the home office, scheduling jobs, billing, resolving issues, and so on.**
- 1 Prepare the first-stage allocation of costs to the activity cost pools.**
 - 2 Compute the activity rates for the activity cost pools.**
 - 3 The company recently completed a 6 hundred square-foot carpet-cleaning job at the Lazy Bee Ranch-a 52-mile round-trip journey from the company's offices in Eagle-Vail. Compute the cost of this job using the activity-based costing system.**
 - 4 The revenue from the Lazy Bee Ranch was \$137.70 (6 hundred square-feet at \$22.95 per hundred square feet). Compute the margin from this job.**
 - 5 What do you conclude concerning the profitability of the Lazy Bee Ranch job? Explain**
 - 6 What advice would you give the president concerning pricing jobs in the future?**

1 Prepare the first-stage allocation of costs to the activity cost pools.

	Distribution of Resource Consumption				
	Across Activity Cost Pools				
	Cleaning Carpets	Travel to Jobs	Job Support	Other	Total
Wages	\$105000	21000	0	14000	140000
Cleaning supplies	25000	0	0	0	25000
Cleaning equipment depreciation	7000	0	0	3000	10000
Vehicle expenses	0	24000	0	6000	30000
Office expenses	0	0	36000	24000	60000
President's compensation	0	0	22500	52500	75000
Total	137000	45000	58500	99500	340000

2 Compute the activity rates for the activity cost pools.

Activity Cost Pool	Activity for the Year	Activity Rates
Cleaning carpets	10,000hundred square feet	\$13.7 per hundred square feet
Travel to jobs	50,000 miles	\$0.9 per mile
Job support	1,800 jobs	\$32.5 per job
Other (costs of idle capacity and organization-sustaining costs)	Not applicable	----

3 The company recently completed a 6 hundred square-foot carpet-cleaning job at the Lazy Bee Ranch-a 52-mile round-trip journey from the company's offices in Eagle-Vail. Compute the cost of this job using the activity-based costing system.

$$6*13.7+52*0.9+32.5*1=\$161.5$$

4 The revenue from the Lazy Bee Ranch was \$137.70 (6 hundred square-feet at \$22.95 per hundred square feet). Prepare a report showing the margin from this job.

$$137.70-161.5=\$-23.8$$

End of Chapter 8

